

Estimation Theory & Regression Inference

Phase 2: A From-Scratch Statistical Engine

Reliability Engineering · Process Regression
Simulation · Confidence Intervals · Prediction

Sharif University of Technology

Department of Industrial Engineering

Engineering Statistics

AUTHOR

Mahdi Abazari Aghdam

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Abstract

This project implements statistical estimation and regression directly from course formulas on two supplied process datasets. For 60 failure times, the method-of-moments and exponential maximum-likelihood estimates both give a mean lifetime of 356.950 hours. A 10,000-trial simulation confirms negligible bias and variance close to the Cramér–Rao bound. The exact 95% confidence interval is [281.411, 467.760] hours; a percentile bootstrap interval is [261.169, 476.962]. In 2,000 repeated datasets, exact and bootstrap intervals cover the plug-in generating value in 0.9510 and 0.9375 of trials. Estimated survival to 500 hours is 0.246. For 40 heat-treatment observations, a directly calculated least-squares fit gives slope 0.73332 HV/°C and $R^2 = 0.93380$. A Gaussian likelihood optimizer independently reproduces the fitted coefficients. At 400°C, predicted mean hardness is 401.929 HV; the interval for one new part is much wider than the interval for the mean. All numbers and figures are regenerated by a fixed-seed Python command. The supplied files are realistic-looking assignment data, so physical generalization requires external validation.

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1 Introduction

The original Phase 2 assignment [1] connects two core ideas: estimating one distribution parameter from component lifetimes, and estimating a functional relation between temperature and hardness. Both can be framed through likelihood, yet their finite-sample uncertainty is different. Part A assumes independent exponential lifetimes with mean θ; Part B assumes a linear mean response with independent Gaussian errors. The project tests those assumptions where possible and distinguishes mathematical derivations, numerical answers on the supplied files, and simulation or independent-library verification. The data are provided as realistic-looking course examples; no independent provenance or production sampling design is documented.

2 Methods and reproducibility

Every reported point estimator and regression coefficient is implemented from direct sums in `src/estimation.py` or `src/regression.py`. Distribution quantiles, random number generation, and generic numerical optimization are allowed helpers. Fitting-library equivalents occur only in functions with a `_verify` suffix and do not feed the reported answers. A local NumPy generator uses seed 403109076. The bias study uses 10000 trials; the consistency study uses 3000 trials at each of five sample sizes. Dataset bootstrap intervals use 5000 resamples. The coverage study uses 2000 repeated datasets and 1000 bootstrap resamples per dataset. The latter count is lower to keep the full coverage experiment computationally practical. A separate regression simulation uses 10000 trials. All 14 figures are exported at 300 dpi. Confidence intervals use a two-sided 95% level unless stated otherwise. The exact exponential interval has model-based coverage; a bootstrap interval has approximate coverage and depends on the empirical distribution [2].

3 Part A: Exponential reliability estimation

3.1 A1. Exploratory description and model plausibility

The 60 observed lifetimes have mean 356.950 hours, median 237.400, sample SD 429.883, minimum 6.800, and maximum 2468.600. Their sample coefficient of variation is $s/\bar{x} = 1.204$. For an Exponential(θ) distribution, both the population mean and standard deviation equal θ, so its population CV is exactly one. The observed CV is moderately above one, and the long upper tail in Figure 1 makes the model plausible as a working approximation, not proven. A single large lifetime can materially affect both mean and CV.

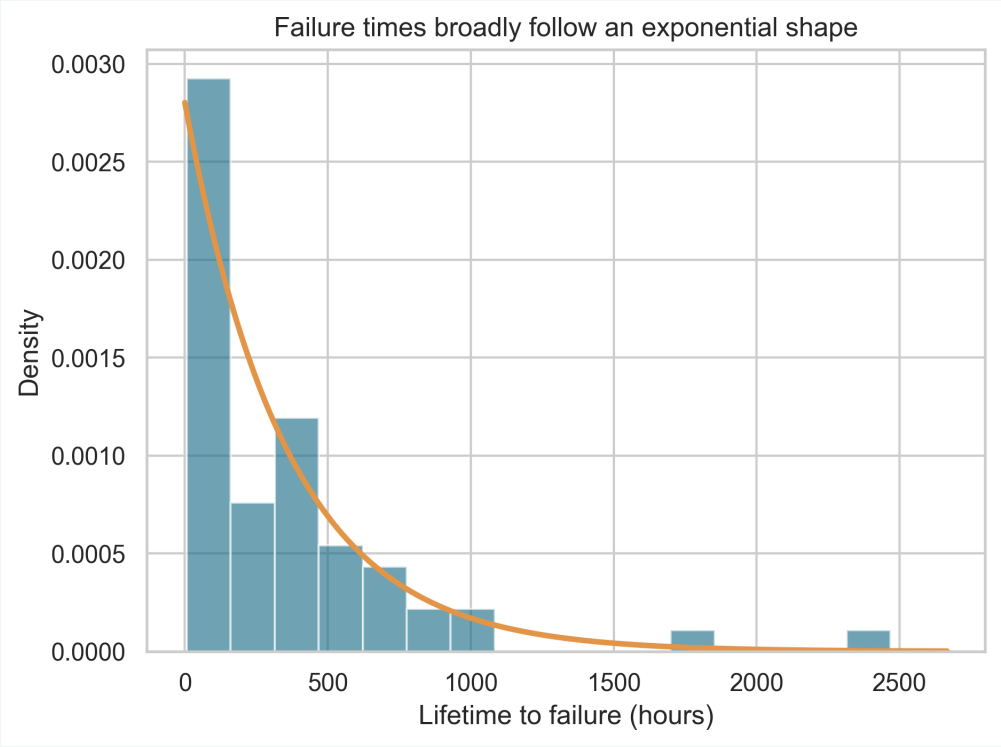


Figure 1: The 60 supplied lifetimes are right-skewed; the fitted exponential density is overlaid for visual assessment.

3.2 A2. Method-of-moments estimate

Equating the first sample moment to $E[X] = \theta$ gives $\hat{\theta}_{\text{MoM}} = \bar{X} = \frac{1}{n}\sum_i X_i$. Direct summation yields 356.950 hours.

3.3 A3. Closed-form and numerical maximum likelihood

The log likelihood is $\ell(\theta) = -n\log\theta - \sum_i x_i/\theta$. Its score vanishes at $\hat{\theta}_{\text{MLE}} = \bar{x} = 356.950$ hours. The second derivative at that point is negative, as shown in Appendix 6. The MoM and MLE answers coincide because the sample mean is both the first empirical moment and the statistic through which the exponential scale enters the likelihood. A separate generic optimizer returns 356.950 hours, and a separately named SciPy fit agrees to numerical precision. Neither verification function supplies the answer.

3.4 A4. Bias and variance, proven and simulated

Linearity of expectation gives $E[\bar{X}] = \theta$; independence gives $\mathrm{Var}(\bar{X}) = \theta^2/n$. In 10000 repeated 60-item samples generated at plug-in $\theta = 356.950$ hours, the mean estimate is 356.683 hours and empirical bias is -0.267 hours. Empirical variance is 2084.362 hours^2, compared with theoretical 2123.555 hours^2. Small differences are Monte Carlo noise, not estimator bias.

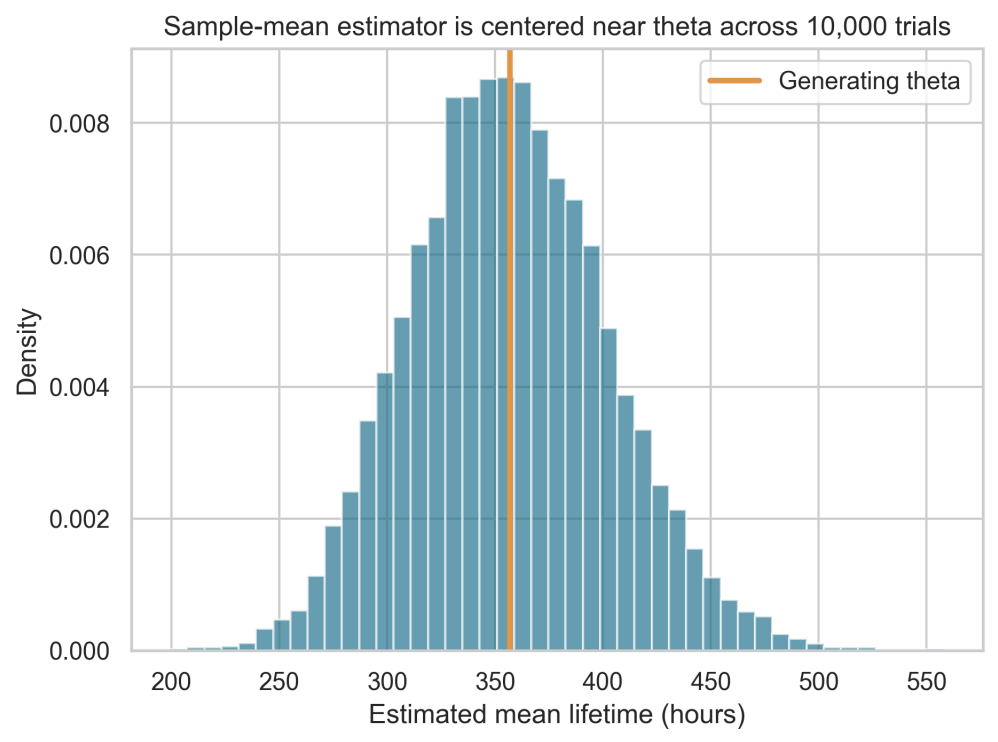


Figure 2: Across 10,000 independent samples of size 60, estimated mean lifetimes cluster near the generating value.

3.5 A5. Consistency and large-sample shape

Because the estimator is unbiased and $\mathrm{Var}(\bar{X}) = \theta^2/n \to 0$, Chebyshev's inequality implies $\bar{X} \xrightarrow{p} \theta$. Table 1 compares empirical and theoretical variance across five n values, using 3000 replications per row. Figures 3 and 4 show the expected $1/n$ decline. At $n = 5000$, standardized estimates have Q–Q $R^2 = 0.999$, consistent with the central limit theorem.

Table 1: Monte Carlo variance of $\hat{\theta}$ approaches θ^2/n; 3,000 trials per sample size.

n	Empirical variance (hours^2)	Theoretical variance (hours^2)
10	12771.634	12741.330
50	2503.263	2548.266
200	650.906	637.067
1000	124.364	127.413
5000	25.365	25.483

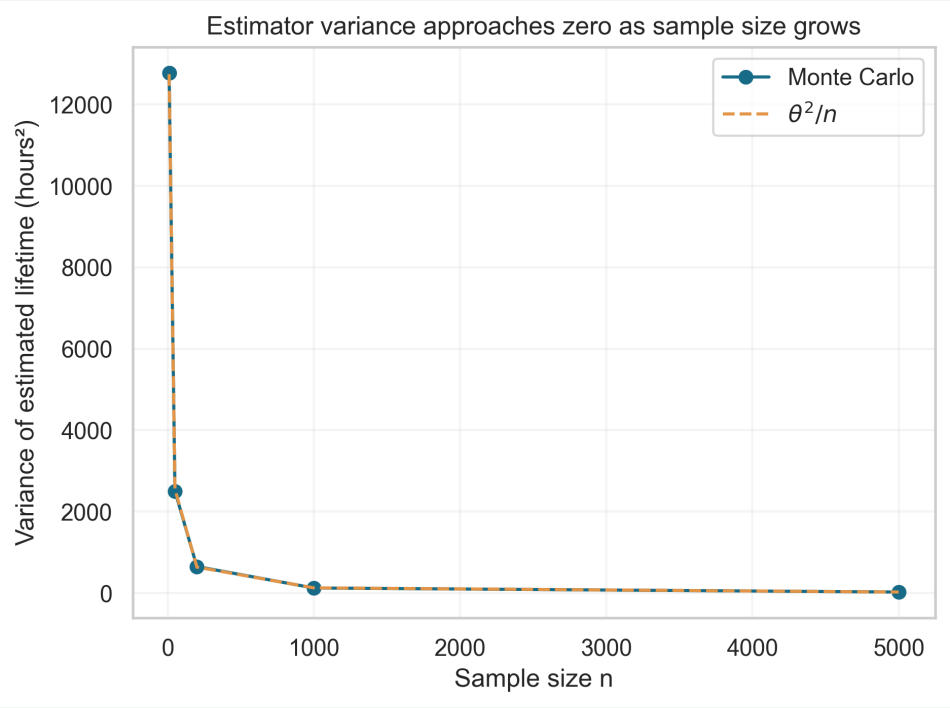


Figure 3: Empirical estimator variance follows the theoretical θ^2/n curve.

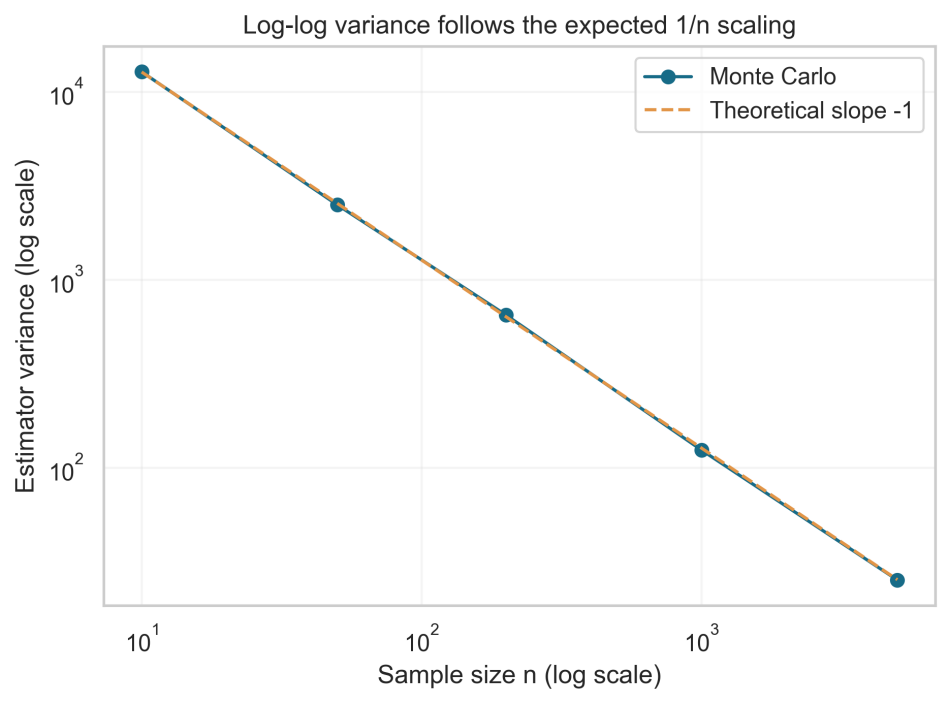


Figure 4: On log–log axes, estimator variance follows approximately a slope of -1.

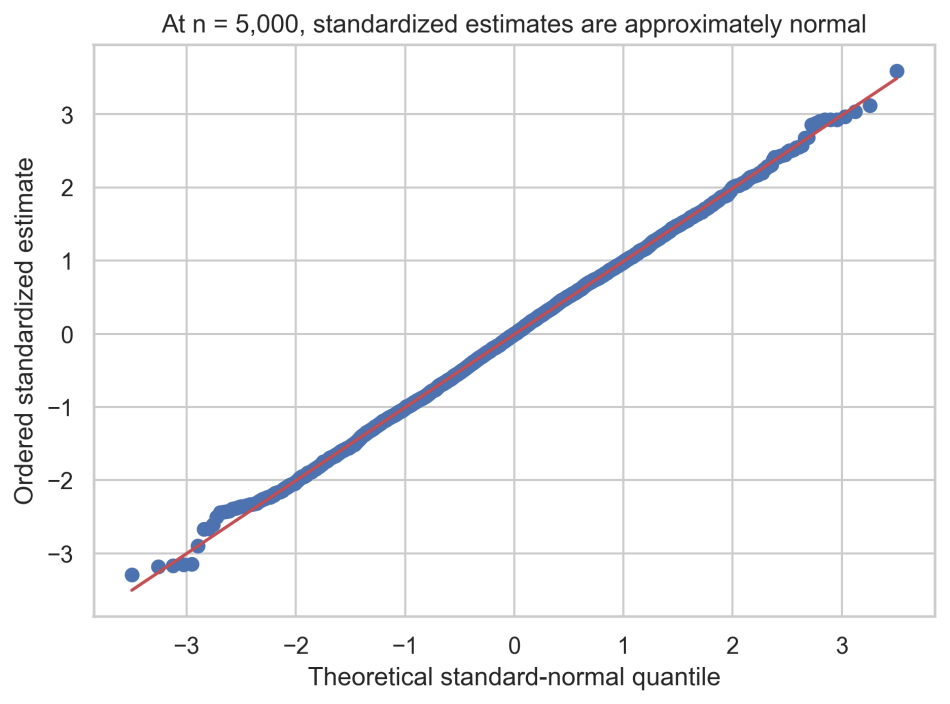


Figure 5: At $n = 5000$, standardized Monte Carlo estimates are approximately normal.

3.6 A6. Cramér–Rao bound and efficiency

One exponential observation has Fisher information $1/\theta^2$, so n independent observations have information n/θ^2. The variance lower bound for an unbiased regular estimator is therefore $\theta^2/n = 2123.555$ hours^2. The sample mean has exactly this theoretical variance and is efficient under the assumed model. The numerical ratio $\mathrm{CRLB}/\widehat{\mathrm{Var}}_{MC} = 1.019$ is slightly above one because the empirical variance is itself noisy; it does not contradict the bound.

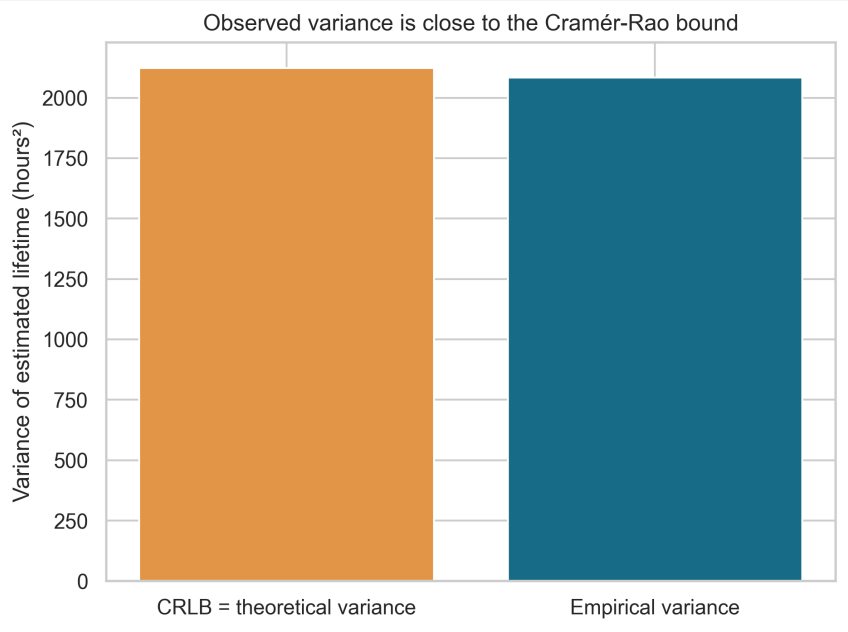


Figure 6: The 10,000-trial empirical variance is close to the Cramér–Rao bound.

3.7 A7. Exact versus bootstrap confidence intervals

The pivot $2n\bar{X}/\theta \sim \chi^2_{2n}$ gives an exact 95% interval of [281.411, 467.760] hours (width 186.349). A 5000-resample nonparametric percentile interval is [261.169, 476.962] hours (width 215.793). The bootstrap is wider in this observed sample. It can be useful when no pivotal formula is available, but it does not inherit exact coverage from the exponential model.

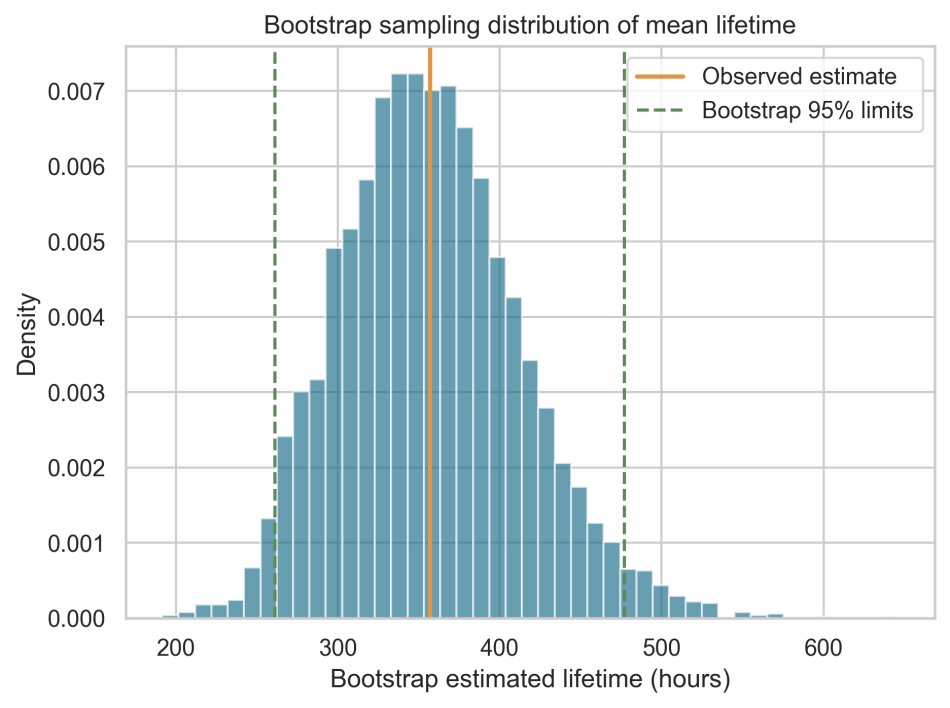


Figure 7: The 5,000-resample distribution of mean lifetime shows the percentile interval around the observed estimate.

In 2000 repeated 60-item datasets generated at the plug-in value, the exact interval covered the generating parameter in 0.9510 of trials and the bootstrap interval in 0.9375. The latter used 1000 resamples per dataset. Figure 8 compares both to the nominal .95 target. This experiment checks the procedures under an assumed exponential generator; it does not establish that the real production mechanism is exponential.

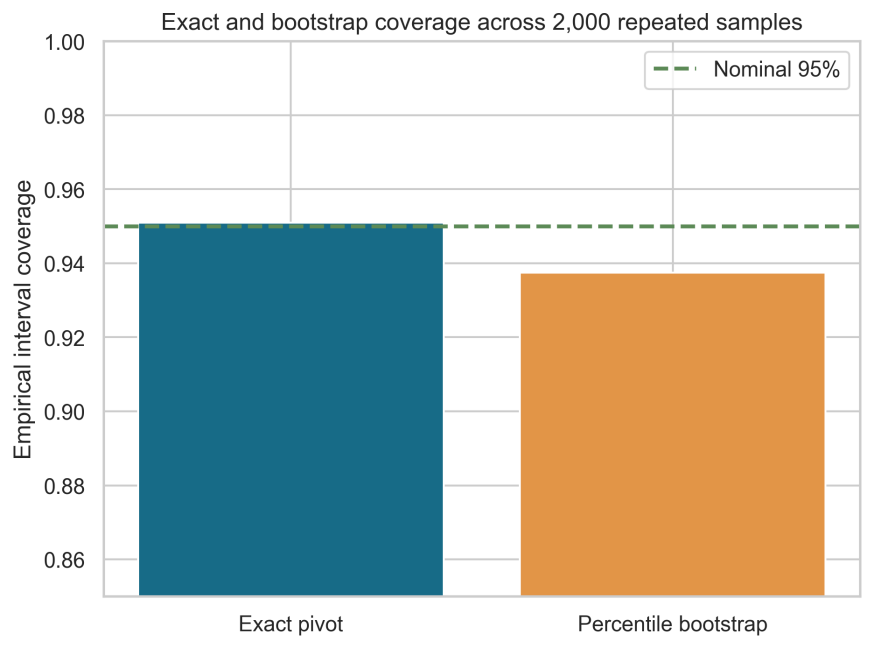


Figure 8: Exact coverage is near 95%; percentile bootstrap coverage is modestly lower across 2,000 model-generated samples.

3.8 A8. Reliability at 500 hours

The exponential survival function is

$$R(t) = \exp(-t/\theta), \qquad \hat{R}(500) = \exp(-500/\hat{\theta}) = 0.246.$$

The estimate follows from MLE invariance. Transforming each of the 5000 bootstrap mean estimates gives a 95% interval [0.147, 0.351]. This is a model-based survival probability at 500 hours, not a directly observed proportion of components in the supplied sample.

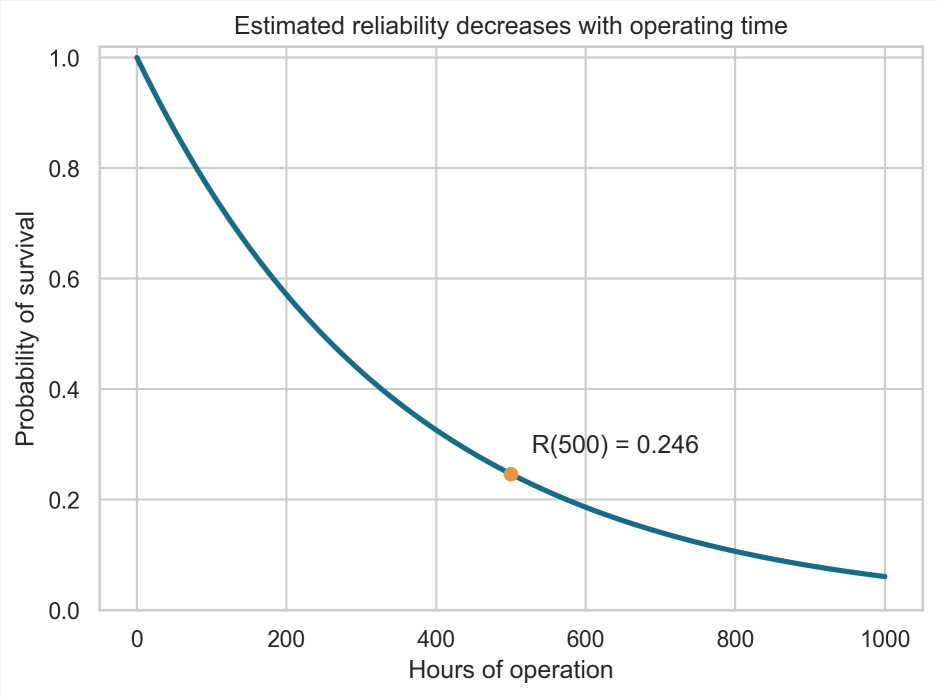


Figure 9: Estimated survival probability declines with operating time; the 500-hour estimate is highlighted.

4 Part B: Temperature and hardness regression

4.1 B1. Scatterplot and linearity

The file contains 40 pairs spanning 172.000–553.000°C. Figure 10 shows a strong approximately linear increase in hardness over this observed range. Curvature or fanning in a residual plot would weaken that description. The file documents no randomization of treatment temperature, so the fitted association should not be called a causal effect without further process-design evidence.

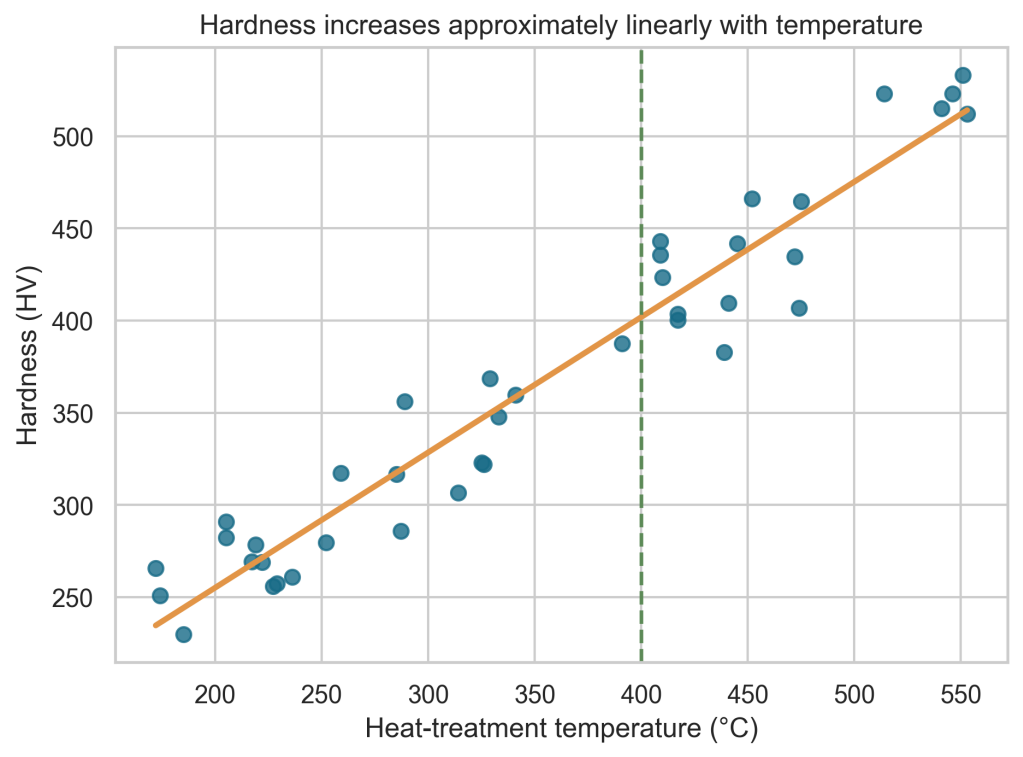


Figure 10: A line fitted from direct sums describes the temperature–hardness trend across 40 observations.

4.2 B2. Least-squares fit from direct sums

The course formulas give $S_{xx} = \sum(x_i - \bar{x})^2 = 545178.775$ and $S_{xy} = \sum(x_i - \bar{x})(y_i - \bar{y}) = 399788.925$. Thus $b = S_{xy}/S_{xx} = 0.73332$ HV/°C and $a = \bar{y} - b\bar{x} = 108.602$ HV. The fitted relation is $\hat{y} = 108.602 + 0.73332x$. An independent SciPy linear fit agrees to numerical precision; it is only a verification.

4.3 B3. Equivalence of least squares and Gaussian MLE

With Gaussian errors, $\ell(A, B, \sigma^2) = -\frac{n}{2}\log(2\pi\sigma^2) - SSE(A, B)/(2\sigma^2)$. At any positive variance, maximizing over A, B is equivalent to minimizing SSE. Differentiation yields the same two normal equations as least squares (Appendix 6). A generic numerical likelihood optimizer returns intercept 108.602 and slope 0.73332, agreeing with the direct fit to 10^{-5} or better. The coefficient equivalence does not mean all variance estimates coincide: the Gaussian MLE uses SSE/n.

4.4 B4. Error variance

The direct residual sum of squares is $SSE = 20782.295$ HV^2. Dividing by $n - 2 = 38$ gives unbiased $s^2 = 546.902$ HV^2, and residual standard error $s = 23.386$ HV. This is a typical vertical deviation of observed hardness from the fitted line. The Gaussian variance MLE is 519.557 HV^2, smaller because its divisor is n.

4.5 B5. Intervals and slope hypothesis test

Using t_{38} with critical value 2.024, the 95% intercept interval is [84.965, 132.239] HV and the 95% slope interval is [0.66920, 0.79744] HV/°C. The test $H_0 : B = 0$ against $H_1 : B \neq 0$ gives $t = 23.153$, $df = 38$, and $p = 5.24 \times 10^{-24}$. The data therefore support a nonzero linear association at $\alpha = .05$. Since 0°C is outside the observed range, the intercept is a model parameter rather than an operating-point hardness prediction. In 10000 new 40-part datasets generated from the fitted Gaussian line, direct-sum slope estimates show bias -4.34×10^{-4} HV/°C. Their empirical variance is 9.94×10^{-4}, compared with theoretical $s^2/S_{xx} = 0.00100$. The 95% slope interval covered the generating slope in 0.95250 of trials. These checks support the formula implementation under the plug-in model, without proving that the factory errors are Gaussian.

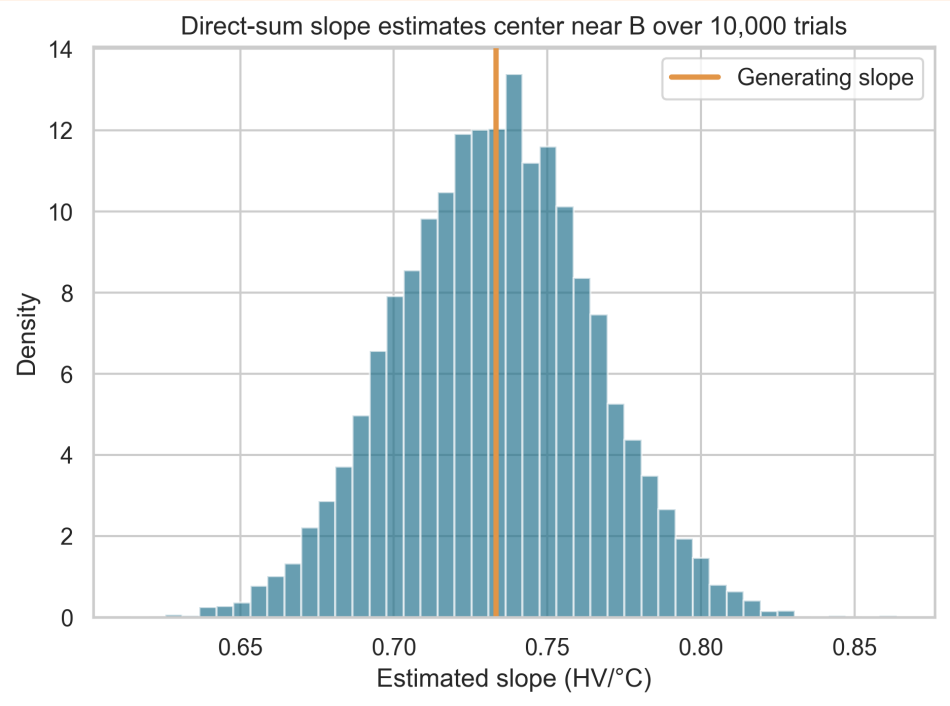


Figure 11: Across 10,000 plug-in Gaussian samples, direct-sum slopes cluster around the generating coefficient.

4.6 B6. Coefficient of determination

With $S_{yy} = 313954.355$ HV^2, $R^2 = 1 - SSE/S_{yy} = 0.93380$. Thus the fitted line accounts for about 93.4% of the hardness variation *within this sample*. High R^2 alone does not validate the Gaussian or constant-variance assumptions.

4.7 B7. Residual diagnostics

Figure 12 plots residuals against fitted hardness; Figure 13 checks normal shape. Shapiro–Wilk yields $W = 0.971$, $p = 0.391$; no clear normality departure is detected at .05. The Breusch–Pagan auxiliary regression gives $nR^2 = 0.546$, $p = 0.460$; no clear linear variance trend with temperature is detected. These non-rejections are not proofs of the assumptions. Durbin–Watson is 1.593 in the supplied row order. Since rows are temperature sorted rather than time ordered, that statistic has no valid temporal-autocorrelation interpretation here.

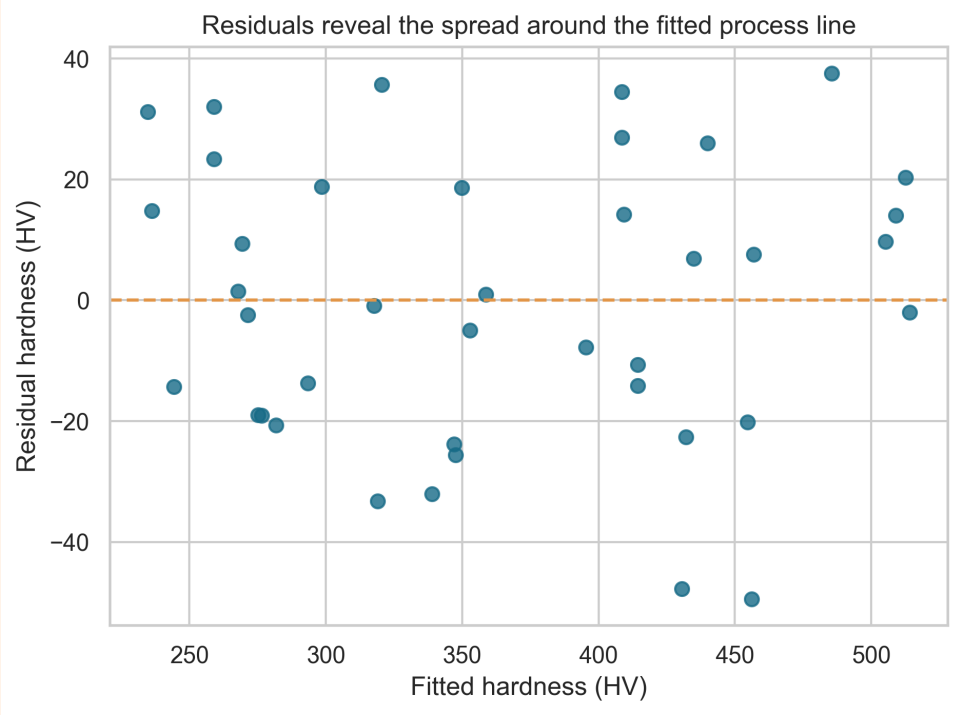


Figure 12: Residuals versus fitted hardness provide a visual check for curvature and changing spread.

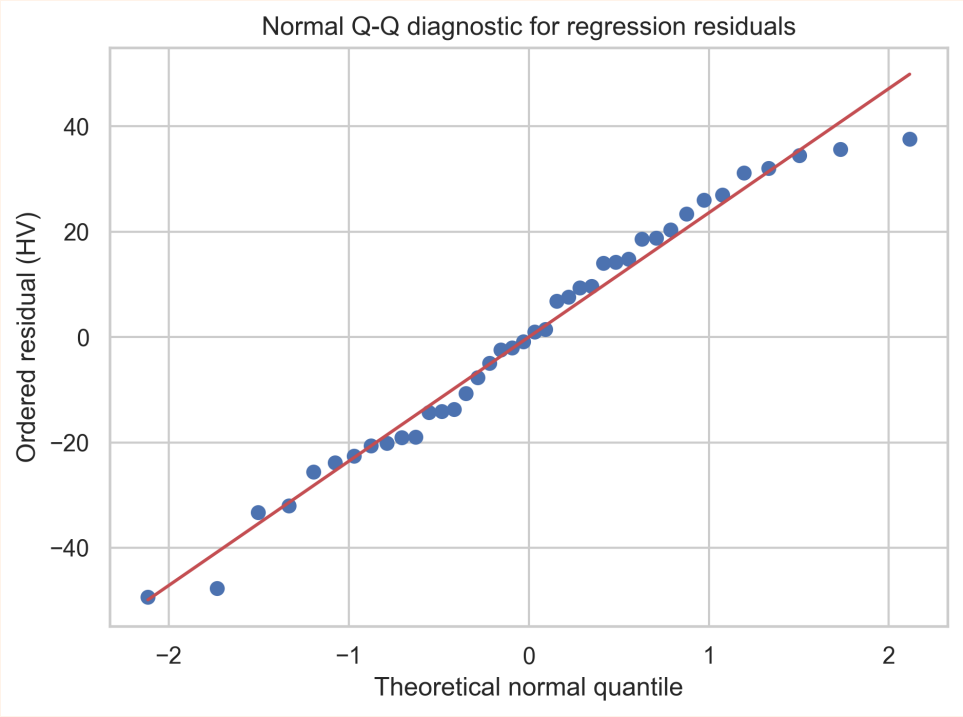


Figure 13: The residual normal Q–Q plot shows no strong departure from a straight-line pattern.

4.8 B8. Mean response and one-part prediction at 400°C

The fitted mean at $x_0 = 400$°C is $\hat{y}_0 = 401.929$ HV. The 95% confidence interval for the *process mean* is [393.778, 410.081] HV. The 95% prediction interval for the hardness of *one new part* is [353.890, 449.968] HV. The latter is wider because it includes individual residual variation as well as uncertainty in the fitted mean. A production engineer estimating average process hardness should use the mean interval; an engineer assessing one upcoming part should use the prediction interval. This is interpolation because 400°C falls within the observed temperature range.

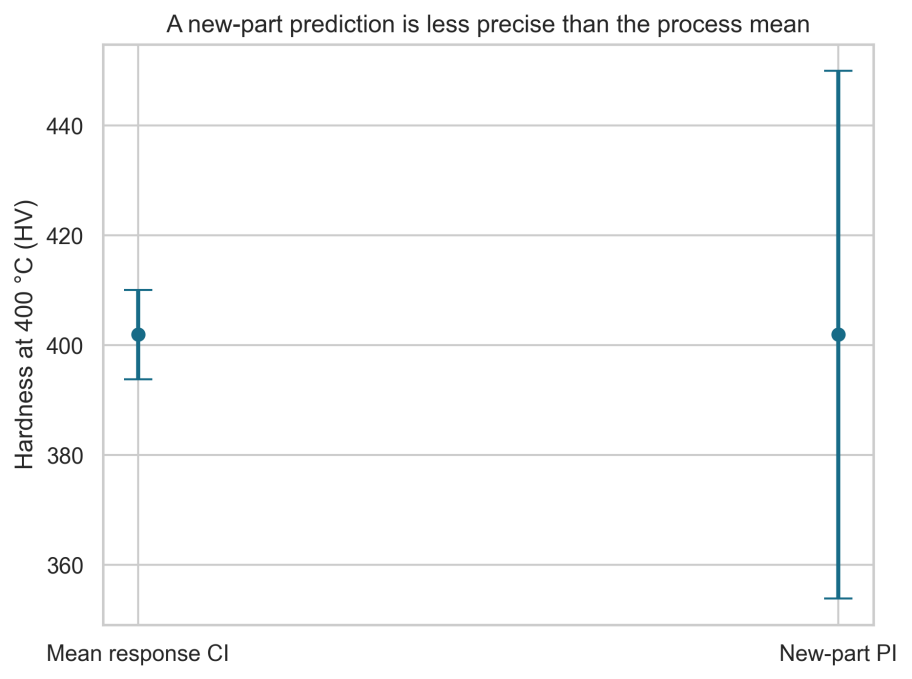


Figure 14: At 400°C, the interval for a new part is wider than the mean-response interval.

5 Cross-case discussion and conclusions

Part A's MoM–MLE equality and Part B's LSE–MLE equality are two instances of optimizing a parametric model through sufficient summary quantities. In Part A the sample mean determines the exponential likelihood's optimum; in Part B the residual sum of squares determines the Gaussian coefficient optimum. Analytical statements and Monte Carlo checks play different roles: proof establishes the property under assumptions, while simulation exposes implementation mistakes and finite-trial variability. The efficiency ratio slightly above one illustrates that a finite Monte Carlo variance estimate is not a mathematical variance bound. The exact interval's observed coverage near .95 supports its implementation under the assumed generator; the bootstrap's lower coverage shows why an approximate interval should be checked in skewed samples. Neither check validates the physical model without external production evidence.

- Mean component lifetime is estimated at 356.950 hours by both MoM and MLE; the observed CV of 1.204 makes the exponential assumption plausible but uncertain.
- The sample mean is unbiased and theoretically reaches the Cramér–Rao bound under the exponential model; 10,000 simulations agree within Monte Carlo error.
- Exact and bootstrap 95% intervals differ in width and empirical coverage; the estimated 500-hour survival probability is 0.246.
- Hardness rises by about 0.73332 HV per 1°C in the fitted range; the slope interval excludes zero.
- A new-part prediction at 400°C is materially less precise than the estimated process mean at that temperature.

6 Appendix: mathematical derivations

Exponential MoM, MLE, information, and exact interval

For $f(x \mid \theta) = \theta^{-1}e^{-x/\theta}$, $x > 0$, $E[X] = \theta$ and $\mathrm{Var}(X) = \theta^2$. Thus the first-moment equation $\bar{X} = \theta$ gives $\hat{\theta}_{MoM} = \bar{X}$. For n independent observations,

$$
\begin{aligned}
L(\theta) &= \theta^{-n}\exp\!\left(-\frac{\sum_i x_i}{\theta}\right),\\
\ell(\theta) &= -n\log\theta - \frac{\sum_i x_i}{\theta},\\
\ell'(\theta) &= -\frac{n}{\theta} + \frac{\sum_i x_i}{\theta^2} = 0 \quad\Longrightarrow\quad \hat{\theta}_{MLE} = \frac{1}{n}\sum_i x_i = \bar{x},\\
\ell''(\hat{\theta}) &= \frac{n}{\hat{\theta}^2} - \frac{2\sum_i x_i}{\hat{\theta}^3} = -\frac{n}{\hat{\theta}^2} < 0.
\end{aligned}
$$

Unbiasedness follows from $E[\bar{X}] = n^{-1}\sum_i E[X_i] = \theta$; independence gives $\mathrm{Var}(\bar{X}) = n^{-2}\sum_i \theta^2 = \theta^2/n$. For one observation, $-E[\ell_1''(\theta)] = 1/\theta^2$; information adds over independent observations, so $I_n(\theta) = n/\theta^2$ and the Cramér–Rao lower bound is $1/I_n = \theta^2/n$. The exact pivot $2\sum_i X_i/\theta = 2n\bar{X}/\theta \sim \chi^2_{2n}$ implies

$$
P\left(\frac{2n\bar{X}}{\chi^2_{2n,1-\alpha/2}} \le \theta \le \frac{2n\bar{X}}{\chi^2_{2n,\alpha/2}}\right) = 1-\alpha.
$$

Since the survival function is $g(\theta) = e^{-t/\theta}$, MLE invariance gives $\hat{R}(t) = g(\hat{\theta}_{MLE})$.

Regression normal equations and Gaussian likelihood

Let $SSE(A, B) = \sum_i (y_i - A - Bx_i)^2$. Differentiating gives

$$\frac{\partial SSE}{\partial A} = -2\sum_i (y_i - A - Bx_i) = 0,$$
$$\frac{\partial SSE}{\partial B} = -2\sum_i x_i(y_i - A - Bx_i) = 0.$$

The first equation gives $A = \bar{y} - B\bar{x}$. Substitution into the second gives $S_{xy} - BS_{xx} = 0$, so $b = S_{xy}/S_{xx}$ and $a = \bar{y} - b\bar{x}$. Under independent $N(0, \sigma^2)$ errors,

$$\ell(A, B, \sigma^2) = -\frac{n}{2}\log(2\pi\sigma^2) - \frac{SSE(A, B)}{2\sigma^2}.$$

For any $\sigma^2 > 0$, the coefficient score equations are $\partial SSE/\partial A = 0$ and $\partial SSE/\partial B = 0$, proving coefficient MLE equals LSE. Maximizing over variance gives $\hat{\sigma}^2_{MLE} = SSE/n$; correcting degrees of freedom for an unbiased estimate gives $s^2 = SSE/(n-2)$.

Coefficient and response uncertainty

With $s^2 = SSE/(n-2)$, the slope and intercept standard errors are $s_b = s/\sqrt{S_{xx}}$ and $s_a = s\sqrt{1/n + \bar{x}^2/S_{xx}}$. Their 95% intervals are estimate $\pm t_{.975,n-2}$ times the respective standard error. Testing $H_0 : B = 0$ uses $t = b/s_b$. At x_0, mean-response standard error is $s\sqrt{h_0}$ with $h_0 = 1/n + (x_0 - \bar{x})^2/S_{xx}$. One future observation has standard error $s\sqrt{1 + h_0}$, yielding the wider prediction interval.

7 Software and data notes

The analysis was verified with Python 3.13, NumPy 2.2.6, pandas 2.2.3, SciPy 1.17.1, Matplotlib 3.10.8, Seaborn 0.13.2, and nbformat 5.10.4. Exact dependency pins are in `requirements.txt`. The original assignment supplied both CSV files and specified the model assumptions [1]. The report and notebook preserve those assumptions explicitly rather than representing the files as independently verified factory measurements.

References

[1] Sharif University of Technology, *Engineering Statistics: Phase 2 Project, Parameter Estimation and Simple Linear Regression*, supplied course assignment.

[2] B. Efron and R. Tibshirani, *An Introduction to the Bootstrap*, Chapman & Hall/CRC, 1993.

[3] D. C. Montgomery and G. C. Runger, *Applied Statistics and Probability for Engineers*, Wiley.